VOYAGEAI PRO

Software Requirements Specification

AI-Powered Freight Forecasting & Vessel Chartering Decision Support Platform

Version 1.0
Prepared: August 2026
Conforms to the approved 6-layer VoyageAI Pro system architecture

> *This SRS is written against the locked architecture: User Channels → Intake & Gateway → Application & Processing → Core AI Decision Engine → Supporting Services → Data Storage. Every functional requirement below traces to exactly one component in that architecture (see Appendix B).*

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for VoyageAI Pro, an AI-powered decision-support platform that transitions maritime vessel chartering from reactive spot-market booking to a proactive, data-driven optimization strategy. It is intended for use by the development team, QA engineers, project stakeholders, and port-owner/chartering-management representatives who will validate the delivered system against these requirements.

1.2 Document Conventions

Requirements are labeled REQ-<Layer>.<Component>.<Number> (e.g., REQ-1.1.2) and traced back to the architecture in Appendix B. The keyword SHALL indicates a mandatory requirement; SHOULD indicates a recommended but non-mandatory requirement; MAY indicates an optional capability.

1.3 Intended Audience and Reading Suggestions

- Developers and architects: read Sections 3 and 4 in full.
- QA/Test engineers: use Section 3's requirement IDs to build test cases; see Appendix B for traceability.
- Project managers / port-owner stakeholders: Sections 1, 2 and 5 give the business-level view.
- Security/compliance reviewers: Section 5.2 and 5.7.

1.4 Project Scope

VoyageAI Pro accepts cargo requirements (type, quantity), origin/destination ports, number of voyages, required delivery period, and contract duration. It analyzes historical freight rates, vessel characteristics, port infrastructure constraints, live congestion, weather, and macroeconomic indices, and returns a recommendation covering: the optimal vessel, the optimal contract type (spot / consecutive-voyage / time charter), the best market-entry window, a risk assessment, an idle-time/repositioning suggestion, and a quantified cost-savings estimate. The system serves port owners, chartering managers, and organization administrators, and integrates with external maritime data feeds (AIS, Baltic Exchange/FFA, weather, port authority congestion data).

1.5 References

- VoyageAI Pro System Architecture & Technical Documentation (approved architecture diagram and component descriptions) — source of truth for this SRS.
- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications (structural reference).

2. Overall Description

2.1 Product Perspective

VoyageAI Pro is a new, standalone cloud-native platform organized into six architectural layers. It

is not a modification of an existing legacy system. The table below summarizes each layer as defined in the approved architecture; all functional requirements in Section 3 are organized against this same structure.

Layer	Components	Role
1. User Channels	Port Owner Dashboard, Chartering Manager Interface, Operational Alerts Service	Human-facing entry points into the system
2. Intake & Gateway	API Gateway, Voyage Parameter Intake	Authenticates, validates and admits every request
3. Application & Processing	Voyage Planning Engine, Market & Congestion Ingestion, Cost & Savings Calculator	Deterministic business logic and data aggregation
4. Core AI Decision Engine	Freight Forecasting, Vessel Matching & Optimization, Market Entry Timing Agent, Risk & Congestion Prediction, Idle-Time & Repositioning Optimizer	AI/ML models producing the recommendation
5. Supporting Services	User & Organization Management, Notification & Alert Dispatcher, Audit/Simulation/Reporting	Cross-cutting platform services
6. Data Storage	PostgreSQL, TimescaleDB/S3, Redis	Persistence and caching

2.2 Product Functions

At a high level, VoyageAI Pro SHALL provide the following functions:

- Accept and validate a chartering request (cargo, route, voyages, delivery window, contract duration).
- Forecast future freight rates for the specified route and cargo type.
- Recommend the most suitable vessel class subject to port/draft constraints.
- Identify the optimal market-entry timing window.
- Compare spot, consecutive-voyage (CVC), and time-charter contract economics.
- Predict vessel idle time and suggest repositioning/backhaul options.
- Score and surface freight, congestion, weather and operational risk.
- Quantify total cost and projected savings versus a spot-market baseline.
- Support multi-tenant organizations with role-based access control.
- Log every AI decision for audit and compliance purposes.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Port Owner	Executive-level user; reviews portfolio savings, risk summaries and voyage tracking; approves high-value recommendations.	Low — needs a summarized, visual interface
Chartering Manager	Primary operational user; submits chartering requests, reviews AI recommendations, runs contract simulations, can override AI parameters.	Medium — domain expert, not necessarily technical
Organization	Manages users, roles, billing and integrations for the	Medium

User Class	Description	Technical Expertise
Administrator	organization (multi-tenant).	
System Administrator	Manages platform-wide configuration, monitors system health, manages model deployments.	High — technical staff
Broker (external, indirect)	Receives finalized chartering terms via notification/webhook; does not log into the platform directly in v1.	N/A

2.4 Operating Environment

- Client: modern evergreen web browsers (Chrome, Edge, Safari, Firefox — last 2 major versions), responsive down to tablet width; mobile-responsive web app for on-the-go access.
- Server: cloud-hosted (AWS/GCP/Azure), containerized services behind an API Gateway.
- Database: PostgreSQL (relational), TimescaleDB or equivalent time-series store, Redis (cache).

2.5 Design and Implementation Constraints

- Authentication SHALL use JWT/OAuth2 as defined at the API Gateway layer.
- All AI/ML outputs SHALL be constrained to a validated schema before being consumed by the workflow/UI layer (no free-form model output reaches the user unvalidated).
- The system SHALL be multi-tenant; one organization's data SHALL NOT be visible to another organization under any circumstance.
- The Core AI Decision Engine SHALL be isolated from the Application layer so individual models (forecasting, vessel matching, timing, risk, idle-time) can be updated or replaced independently.

2.6 Assumptions and Dependencies

- The system depends on the continued availability of third-party data feeds: AIS vessel telemetry, Baltic Exchange/FFA freight indices, port authority congestion data, and weather/meteorological data providers.
- It is assumed that chartering managers have basic familiarity with maritime chartering terminology (spot, CVC, time charter, DWT, draft, laycan).
- Historical freight-rate data of sufficient volume and quality is assumed to be available to train and validate the forecasting models.

3. System Features (Functional Requirements)

3.1 User Channels Layer

1.1 Port Owner Dashboard — Priority: High

A responsive web portal providing executive-level insights, savings analytics, portfolio risk summaries, and voyage tracking.

Req ID	Requirement
REQ-1.1.1	The system SHALL display a summary of active voyages, total savings for a selected period, and current portfolio risk level.
REQ-1.1.2	The system SHALL display a list/timeline of vessels currently idle with predicted idle

Req ID	Requirement
	duration.
REQ-1.1.3	The system SHALL allow the Port Owner to approve or reject a high-value AI recommendation before it is sent to a broker.
REQ-1.1.4	The system SHALL allow filtering the dashboard by date range and by route/vessel.
REQ-1.1.5	The system SHALL allow exporting a summary report as PDF.

1.2 Chartering Manager Interface — Priority: High

A comprehensive management interface providing granular control over route planning, contract simulation comparisons, and manual parameter overrides.

Req ID	Requirement
REQ-1.2.1	The system SHALL allow the Chartering Manager to create, view, and edit chartering requests.
REQ-1.2.2	The system SHALL allow side-by-side comparison of spot vs. consecutive-voyage vs. time-charter contract simulations.
REQ-1.2.3	The system SHALL allow manual override of any AI-suggested parameter (vessel class, timing window, contract type) with a mandatory justification note.
REQ-1.2.4	The system SHALL persist a full history of manual overrides against the original AI recommendation for audit purposes.

1.3 Operational Alerts Service — Priority: High

Automated dispatch system delivering urgent notifications via Email, SMS, or Webhooks.

Req ID	Requirement
REQ-1.3.1	The system SHALL dispatch an alert within 60 seconds of a qualifying trigger (e.g., freight volatility spike, port congestion risk, SLA breach).
REQ-1.3.2	The system SHALL support Email, SMS, and Webhook as notification channels, configurable per user and per alert type.
REQ-1.3.3	The system SHALL allow users to acknowledge an alert, and SHALL record the acknowledgement timestamp and user.

3.2 Intake & Gateway Layer

2.1 API Gateway — Priority: High

Central ingress point enforcing authentication, rate limiting, and request sanitization.

Req ID	Requirement
REQ-2.1.1	The system SHALL authenticate every incoming request using JWT/OAuth2 and reject unauthenticated requests with HTTP 401.
REQ-2.1.2	The system SHALL enforce per-user and per-organization rate limits and reject excess requests with HTTP 429.
REQ-2.1.3	The system SHALL sanitize and validate all inbound payloads against a defined schema before forwarding to internal services.

2.2 Voyage Parameter Intake — Priority: High

Parses and validates multi-voyage operational requirements.

Req ID	Requirement
REQ-2.2.1	The system SHALL require cargo type, cargo quantity (MT), origin port, destination port, number of voyages, and delivery window as mandatory fields on every chartering request.
REQ-2.2.2	The system SHALL validate that the origin and destination ports exist in the port reference dataset before accepting the request.
REQ-2.2.3	The system SHALL reject a request with a clear validation message if the cargo quantity is incompatible with the minimum/maximum capacity of any vessel class serving that route.

3.3 Application & Processing Layer

3.1 Voyage Planning Engine — Priority: High

Computes optimal maritime transit routes and validates vessel draft against port restrictions.

Req ID	Requirement
REQ-3.1.1	The system SHALL compute at least one feasible route between the specified origin and destination for every valid request.
REQ-3.1.2	The system SHALL validate candidate vessel draft against the destination port's maximum draft and flag any incompatibility.
REQ-3.1.3	The system SHALL check berth depth and cargo-handling capacity at both origin and destination ports.

3.2 Market & Congestion Ingestion — Priority: High

Continuously aggregates real-time AIS vessel telemetry, port waiting queues, weather data, and macroeconomic indices.

Req ID	Requirement
REQ-3.2.1	The system SHALL refresh AIS vessel telemetry data at least every 15 minutes.
REQ-3.2.2	The system SHALL refresh port congestion/waiting-queue data at least every 30 minutes.
REQ-3.2.3	The system SHALL ingest daily freight index (e.g., Baltic Dry Index / FFA) updates.

3.3 Cost & Savings Calculator — Priority: High

Runs quantitative simulations comparing total cost of ownership across contract types.

Req ID	Requirement
REQ-3.3.1	The system SHALL calculate total projected cost for spot, consecutive-voyage, and time-charter options for a given request.
REQ-3.3.2	The system SHALL calculate and display the projected savings of the recommended option versus a spot-market baseline.
REQ-3.3.3	The system SHALL allow the user to adjust key assumptions (fuel price, freight rate) and re-run the simulation.

3.4 Core AI Decision Engine

4.1 Freight Forecasting Model — Priority: High

Time-series deep learning model forecasting spot and forward freight agreement (FFA) rate trajectories.

Req ID	Requirement
REQ-4.1.1	The system SHALL forecast freight rates for the requested route and cargo type for at least a 90-day horizon.
REQ-4.1.2	The system SHALL return a confidence score (0–100%) alongside every forecast.
REQ-4.1.3	The system SHALL flag a forecast as low-confidence and route it for manual review when confidence falls below a configurable threshold.

4.2 Vessel Matching & Optimization — Priority: High

Uses mixed-integer linear programming (MILP) to pair cargo quantities with optimal vessel classes subject to port compatibility.

Req ID	Requirement
REQ-4.2.1	The system SHALL rank candidate vessel classes (e.g., Capesize, Panamax, Supramax) by fit for the given cargo quantity and port constraints.
REQ-4.2.2	The system SHALL exclude any vessel class that violates a hard port/draft constraint from the recommendation.
REQ-4.2.3	The system SHALL present at least the top 3 ranked vessel options when confidence in the top choice is below the high-confidence threshold.

4.3 Market Entry Timing Agent — Priority: Medium

Reinforcement-learning model evaluating market momentum to pinpoint optimal contract execution windows.

Req ID	Requirement
REQ-4.3.1	The system SHALL output a recommended market-entry date range for every chartering request.
REQ-4.3.2	The system SHALL explain, in plain language, the primary factor(s) driving the recommended timing window.

4.4 Risk & Congestion Prediction — Priority: High

NLP models extracting risk factors from maritime intelligence, bunker price fluctuations, and geopolitical news while predicting port delays.

Req ID	Requirement
REQ-4.4.1	The system SHALL assign a risk level (Low / Medium / High / Critical) to every chartering request.
REQ-4.4.2	The system SHALL list the specific contributing factors behind any Medium-or-higher risk rating.
REQ-4.4.3	The system SHALL update the risk level automatically if a material new risk signal is detected before the contract is finalized, and SHALL notify the assigned Chartering Manager.

4.5 Idle-Time & Repositioning Optimizer — Priority: Medium

Predicts ballast/idle duration post-discharge and suggests backhaul or repositioning voyages.

Req ID	Requirement
REQ-4.5.1	The system SHALL predict expected idle days for the assigned vessel following cargo discharge.
REQ-4.5.2	The system SHALL suggest at least one backhaul or repositioning opportunity when predicted idle time exceeds a configurable threshold.

3.5 Supporting Services Layer

5.1 User & Organization Management — Priority: High

Handles multi-tenancy, enterprise role-based access control (RBAC), and access permissions.

Req ID	Requirement
REQ-5.1.1	The system SHALL support at least the roles: Port Owner, Chartering Manager, Organization Admin, and System Admin.
REQ-5.1.2	The system SHALL enforce that a user can only access data belonging to their own organization (multi-tenant isolation).
REQ-5.1.3	The system SHALL allow an Organization Admin to invite, deactivate, and change the role of users within their organization.

5.2 Notification & Alert Dispatcher — Priority: High

Routes high-priority events, threshold triggers, and contract reminders to designated personnel.

Req ID	Requirement
REQ-5.2.1	The system SHALL allow configuration of notification rules per event type and per role.
REQ-5.2.2	The system SHALL retry failed notification delivery at least twice before marking it as failed and logging the failure.

5.3 Audit, Simulation & Reporting — Priority: High

Logs model inference rationale, tracks system audits for compliance, and compiles exportable scenario-analysis reports.

Req ID	Requirement
REQ-5.3.1	The system SHALL log, for every AI recommendation issued, the model name/version, input snapshot, confidence score, and output.
REQ-5.3.2	The system SHALL provide an exportable (PDF/CSV) report of savings achieved over a selected period.
REQ-5.3.3	The system SHALL retain audit logs for a minimum of 3 years or as required by applicable regulation, whichever is longer.

3.6 Data Storage Layer

6.1 Relational Database (PostgreSQL) — Priority: High

Stores relational schemas including port infrastructure limits, user profiles, voyage history, and

contract records.

Req ID	Requirement
REQ-6.1.1	The system SHALL persist all chartering requests, recommendations, contracts, vessels, ports, and user/organization records in PostgreSQL.
REQ-6.1.2	The system SHALL enforce referential integrity between chartering requests, recommendations, and contracts.

6.2 Time-Series DB / Data Lake — Priority: Medium

High-throughput storage capturing historical Baltic Dry Index points, real-time AIS vessel positions, and weather data.

Req ID	Requirement
REQ-6.2.1	The system SHALL store all ingested freight-index, AIS, and weather data points with a timestamp suitable for time-series querying.
REQ-6.2.2	The system SHALL retain raw historical market data for at least 5 years to support model retraining.

6.3 In-Memory Cache (Redis) — Priority: Medium

Caches high-frequency inference outputs, active session states, and real-time market tickers.

Req ID	Requirement
REQ-6.3.1	The system SHALL cache the most recent AI recommendation per active chartering request to avoid redundant recomputation within a session.
REQ-6.3.2	The system SHALL cache active user session state to support sub-second dashboard response times.

4. External Interface Requirements

4.1 User Interfaces

The primary user interface is a responsive web application covering: Landing/marketing page, Signup, Login, Onboarding, Port Owner Dashboard, Chartering Manager workspace, New Chartering Request form, AI Recommendation results view, Freight Forecast view, Vessel Intelligence, Port Intelligence, Risk & Alerts Center, Reports, and Organization Settings. All screens SHALL follow a single, consistent design system (see the platform's approved UI style guide).

4.2 Hardware Interfaces

No specialized hardware is required. Standard client devices (desktop, laptop, tablet) with a modern web browser and internet connectivity are sufficient. Server-side, the platform runs on standard cloud compute/container infrastructure.

4.3 Software Interfaces

Interface	Purpose

Interface	Purpose
AIS Vessel Telemetry Provider	Live vessel position, speed, heading and status
Baltic Exchange / FFA Data Feed	Freight rate indices used for forecasting
Port Authority / Congestion API	Live berth occupancy and waiting-time data
Weather / Meteorological Data Provider	Route-specific weather and cyclone risk data
Identity Provider (OAuth2/JWT)	Authentication and single sign-on
Email / SMS / Webhook Gateway	Notification and alert delivery
Broker / e-Chartering Platform (optional)	Hand-off of approved chartering terms

4.4 Communication Interfaces

- All client–server communication SHALL use HTTPS (TLS 1.2 or higher).
- Live dashboard updates (vessel position, market ticker) SHOULD use WebSocket or server-sent events for low-latency delivery.
- Outbound integrations (broker hand-off, ERP sync) SHALL support webhook callbacks with signed payloads.

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement	Target
NFR-P1	Dashboard initial load time	≤ 2.5 seconds on broadband connection
NFR-P2	AI recommendation generation time (end-to-end)	≤ 8 seconds for 95% of requests
NFR-P3	Concurrent active users supported (per organization tier)	≥ 200 concurrent users at launch, horizontally scalable
NFR-P4	API Gateway response time (excluding AI computation)	≤ 300 ms at the 95th percentile

5.2 Security Requirements

- The system SHALL encrypt data at rest (AES-256) and in transit (TLS 1.2+).
- The system SHALL implement role-based access control (RBAC) at the API layer, not only in the UI.
- The system SHALL isolate each organization's data such that cross-tenant data access is technically impossible, not merely UI-hidden.
- The system SHALL log all authentication attempts, authorization failures, and administrative actions to an immutable audit trail.
- The system SHALL minimize and redact sensitive commercial data before sending it to any third-party AI/model provider.

5.3 Reliability & Availability

- The system SHALL target 99.5% monthly uptime for core dashboard and recommendation services.
- The system SHALL perform automated daily backups of the relational database with a recovery point objective (RPO) of 24 hours.
- The system SHALL degrade gracefully if a third-party data feed (AIS, weather, freight index) is unavailable — recommendations SHALL be marked as 'reduced confidence' rather than failing outright.

5.4 Scalability

- The system architecture SHALL support independent horizontal scaling of the Core AI Decision Engine separately from the Application layer.
- The system SHALL support onboarding additional organizations (tenants) without downtime.

5.5 Usability

- The system SHALL be usable by a Chartering Manager with no prior training beyond a single onboarding walkthrough.
- The system SHALL be responsive down to tablet screen widths (≥768px).
- The system SHOULD conform to WCAG 2.1 AA accessibility guidelines for core workflows.

5.6 Maintainability & Portability

- Each of the six architectural layers SHALL be independently deployable and independently versioned.
- The system SHALL avoid cloud-provider lock-in for core compute where practical, to support AWS/GCP/Azure portability.

5.7 Compliance & Legal

- The system SHALL support data residency configuration where required by a customer's jurisdiction.
- The system SHALL retain financial/contract records for the period required by applicable maritime commerce and tax regulation in the customer's jurisdiction.

6. Data Requirements

Core entities persisted by the system include: organizations, users, chartering_requests, vessels, ports, freight_rate_history, recommendations, contracts, risk_alerts, idle_time_log, and audit_logs. Raw historical market data (AIS positions, freight indices, weather) is retained in the time-series store separately from transactional records. All entity relationships SHALL be maintained as described in the approved system architecture's data layer.

- Chartering request data SHALL be retained for the lifetime of the customer relationship plus 3 years.
- AI decision audit logs (model, input, confidence, output) SHALL be retained for a minimum of 3 years.
- Raw historical market data SHALL be retained for a minimum of 5 years to support model retraining and backtesting.

7. Appendices

A. Appendix A — Glossary

Term	Definition
CVC	Consecutive Voyage Contract — an agreement covering multiple sequential voyages
DWT	Deadweight Tonnage — a vessel's carrying capacity
FFA	Forward Freight Agreement — a derivative contract on future freight rates
AIS	Automatic Identification System — vessel tracking telemetry
Laycan	The agreed window in which a vessel must be ready to load
Spot Fixture	A single-voyage vessel charter booked at the prevailing market rate
RBAC	Role-Based Access Control

B. Appendix B — Requirement Traceability Matrix

Architecture Layer	Component	Requirement IDs
1. User Channels	Port Owner Dashboard / Chartering Manager Interface / Operational Alerts	REQ-1.1.x / REQ-1.2.x / REQ-1.3.x
2. Intake & Gateway	API Gateway / Voyage Parameter Intake	REQ-2.1.x / REQ-2.2.x
3. Application & Processing	Voyage Planning / Market Ingestion / Cost Calculator	REQ-3.1.x / REQ-3.2.x / REQ-3.3.x
4. Core AI Decision Engine	Forecasting / Vessel Matching / Timing / Risk / Idle-Time	REQ-4.1.x – REQ-4.5.x
5. Supporting Services	User Mgmt / Notifications / Audit & Reporting	REQ-5.1.x / REQ-5.2.x / REQ-5.3.x
6. Data Storage	PostgreSQL / Time-Series DB / Redis	REQ-6.1.x / REQ-6.2.x / REQ-6.3.x

C. Appendix C — Assumptions & Open Issues

- Final list of supported vessel classes and cargo types to be confirmed with pilot customers before general availability.
- Choice of specific third-party AIS/weather/freight-index data providers to be finalized during vendor selection.
- Broker/e-chartering hand-off integration is scoped as a fast-follow, not required for initial launch.